

JAMES ZENG

Software Engineer | Full-Stack Product & AI Systems

+1 (872) 258-8789 | Berkeley, CA (Open to relocate) | [LinkedIn](#) | [GitHub \(link\)](#) | james1.zeng@gmail.com

EDUCATION

University of California, Berkeley - *Master of Analytics*, GPA: 3.88 ([researches](#))

Ohio State University - *B.S. in Statistics; Minor in Computer Science*, GPA 3.81

SKILLS

Machine Learning: Python, Scikit-Learn, XGBoost, PyTorch, TensorFlow, NumPy, pandas, Feature engineering, Model evaluation

Data Engineering: SQL, BigQuery, ClickHouse, ETL pipelines, Dagster, Real-time streaming, Data quality & identity resolution

ML Infrastructure: Docker, Kubernetes, GCP Pub/Sub, AWS Lambda, Redis, CI/CD (GitHub Actions), Model serving

Statistics & Analysis: Statistical modeling, R, Tableau, AB testing, Metrics design, Hypothesis testing

AI & ML Stack: LLM integration, LangChain, RAG, Vector databases, Prompt evaluation

PROFESSIONAL EXPERIENCE

Future Intelligence Labs (AI Automation) - *Artificial Intelligence Engineer* 12/2025 - Present
Berkeley/CA

- Built model evaluation pipeline measuring context quality via RL, improving benchmark accuracy from 80% to 96.7%.
- Trained policy-violation classification model on an 8B Llama detector, evaluating precision and recall for production content safety.
- Built distributed Python pipelines processing 2M+ records, handling missing and sparse features into training-ready datasets.
- Automated ETL and feature pipelines, cutting execution time 70% across iterative model training workloads.
- Orchestrated LangGraph feedback loop on the data collection pipeline, cutting overnight pipeline downtime from 9h to 20min.

Alterion (AI Observability) - *Software Engineer* 10/2025 - 12/2025
Menlo Park/CA

- Built distributed event processing system handling 100K+ events/sec for real-time ML telemetry collection and data pipeline analysis.
- Provisioned scalable data collector/processor pipelines using GCP Pub/Sub, enabling reliable ingestion for downstream ML workflows.
- Designed two-tier pipeline separating real-time ingestion from async processing, reducing end-to-end latency by 45%.
- Resolved production pipeline stability issues within one week, presenting 80% of demos and converting 2 customers.
- Modeled relational schemas for feature correlation analysis supporting 400 QPS and scalable data querying workloads.
- Tuned Kubernetes and Redis caching for identity resolution, recovering 87% of unmatched records through record matching.

Berkeley SkyDeck - *Software Engineer Intern* 04 - 07/2025
Emeryville/CA

- Built *PromptArena* platform for model output evaluation across iterative testing workflows with 200 evaluation iterations monthly.
- Developed deployment framework for 50+ service types; stress-tested 486 concurrent workloads achieving zero failures.
- Eliminated 60s data reload delays and reduced 1M-row query rendering from 15s to ~1s through backend streaming optimization.
- Containerized applications with Docker and automated CI/CD pipelines using GitHub Actions and Jenkins.

EndoVas Medical Inc. - *Software Engineer Intern* 09/2023 - 07/2024
remote & Shanghai/CN

- Developed diagnostic data system using Python and React for remote device monitoring and automated analytics reporting.
- Upgraded serverless analytics pipelines on AWS Lambda, reducing data troubleshooting time from 3 hours to ~30 minutes.
- Conducted 11 on-site client demos, translating analytical findings into operational deployment requirements.

PROJECT EXPERIENCE

Bath-Note Project - *Owner & Developer* 08 - 09/2024
Berkeley/CA

- Developed ML voice-note application with automated classification workflows using Python and NumPy.
- Implemented evaluation-based triggering, reducing LLM token usage by 18% while maintaining output quality.

J.P. Morgan Chase Hackathon - *Lead Developer (1st Place)* 03 - 04/2024
Columbus/OH

- Built XGBoost recommender on BigQuery with real-time analytics, improving MRR by ~35% over baseline models.